

When personalization deters delegation to algorithms: Experimental evidence

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Background

- ▶ Huge & unconnected literature on algorithm/automation/AI/... reliance (→ review by Chugunova & Sele, 2022)
- ▶ Most of the literature: setups with clearly identifiable mistakes
- ▶ But in many real-life settings, preferences determine what is a mistake
- ▶ E.g., robo-advisors: maximal payoff vs best fit to risk-parameters

Relevance?

Well-known (Swiss) robo-advisors(' marketing strategies)

- ▶ “We rely on a low-cost, passive investment strategy and have selected a total of 9 equity, bond and real estate ETFs from over 1,500 ETFs (exchange-traded funds) and put them together in such a way that they **achieve an optimal risk/return ratio.**” (findependent.ch; for GER, see e.g., whitebox.eu)
- ▶ “Chat with Selma: Selma **looks at your financial life and learns how you should structure your finances.** Get your plan: Selma **creates a personalized investment plan that fits your life.** Sit back and relax! Selma takes care of your money from now on and does all the buying and selling for you.” (selma.com; for GER, see e.g., quirion.de)

Research Question

- ▶ Does algorithm reliance differ for **objective optimization vs. preference-based personalisation** in a preferential-choice environment?
 - (i) ...at first contact?
 - (ii) ...overall?

Objective vs Personalised Algorithms

- ▶ What do we mean by **objective optimization**?
 - an algorithm that chooses some ‘objectively ideal’ option independently of a subject’s preferences (here: **highest EV**)
- ▶ What do we mean by **personalised**?
 - an algorithm that measures a subject’s preferences and chooses accordingly (here: **implementation of revealed preferences**)

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 - an algorithm that measures a subject's preferences and chooses accordingly (here: **implementation of revealed preferences**)
- ▶ In the experiment, we describe both algorithms in intuitive terms:

EV: *'The algorithm works out how much each option would yield on average and selects the option for which this is highest.'*

Prefs: *'The algorithm analyses your choices from Part 1 and selects the best option for you in light of your Part-1 choices.'*

Experimental Design

Part 1 – Elicitation of preferences

- ordered lotteries, no foils, unlimited time
- “hypothetical, but the choices may be meaningful for Part 2”

Part 2 – Lottery Selection Task

- could stand for an investment task
- complexity (number of options, outcomes under different SOTW)
- time pressure (think: relative to the number of options)

Tasks

	State 1	State 2	State 3	State 4	State 5	State 6	State 7	State 8	State 9	State 10
Option 1	225	225	225	225	225	225	225	205	0	0
Option 2	525	525	525	525	525	0	0	0	0	0

State:	1	2	3	4	5	6	7	8	9	10
Option 1	225	225	0	225	0	225	225	225	225	0
Option 2	525	0	505	0	525	0	0	0	525	505
Option 3	525	0	525	0	525	0	0	0	525	525
Option 4	225	225	0	225	205	205	205	225	225	0
Option 5	225	205	0	205	205	225	225	205	225	0
Option 6	225	225	0	225	205	225	225	225	225	0
Option 7	525	0	525	0	525	0	0	0	225	525
Option 8	505	0	525	0	505	0	0	0	505	525

► For further distraction: 4 filler tasks in between

► 8 rounds: 5' pre-view — 5' algorithm choice—8' decision/waiting time

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Option 2	525	0	505	0	525	0	0	0	525	505
Option 3	525	0	525	0	525	0	0	0	525	525
Option 4	225	225	0	225	205	205	205	225	225	0
Option 5	225	205	0	205	205	225	225	205	225	0
Option 6	225	225	0	225	205	225	225	225	225	0
Option 7	525	0	525	0	525	0	0	0	225	525
Option 8	505	0	525	0	505	0	0	0	505	525

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Main Results: First Contact

- ▶ **Hypothesis:** Higher delegation to EV-maximizing algorithm than to personalized algorithm.
- ▶ **Finding:** Supported at first contact.
- ▶ Delegation rates in Round 1: **66.9% (EV)** vs. **55.8% (Prefs)**.
- ▶ Boschloo exact test: $p = 0.0413$.
- ▶ Treatment gap is concentrated in the first decision.

Main Results: Repeated Decisions

- ▶ **Average Delegation:** No significant treatment difference overall ($p = 0.2425$).
- ▶ **Dynamics:** Convergence over rounds.
- ▶ Delegation in EV condition declines more strongly after Round 1.
- ▶ **Key Driver:** Negative feedback (specifically prior zero-outcomes) strongly deters subsequent delegation ($\beta = -0.3854, p < 0.001$).

Categorical Non-Delegation

- ▶ A distinct acceptance barrier under personalization.
- ▶ **Never-delegators:**
 - 15.7% in Preference-based condition.
 - 3.4% in EV condition.
- ▶ Not linked to demographics, risk preferences, or AI experience.
- ▶ Not linked to dissatisfaction with Part 1 preferences.

Mechanisms & Interpretations

- ▶ **The "Agency" Trade-off:** Personalization may encroach on users' sense of choice authority.
- ▶ **Deterrence Channels:**
 - *Personalized:* Deterred by expectation of non-preferred choices.
 - *EV:* Deterred by expectation of "completely weird" (extreme) choices.
- ▶ **Conclusion:** Preference-matching can raise initial hesitation and heighten concerns about autonomy, functioning as an acceptance barrier.

Implications for Design

- ▶ Personalization is not a uniformly beneficial acceptance lever.
- ▶ In high-stakes domains (Finance, Healthcare), objective optimization may face lower initial resistance.
- ▶ **Takeaway:** Uptake depends on perceived rule characteristics, not just expected performance.